

Wireshark PCAP Traffic Analysis

Network Forensics Report — traffic.pcapng

Objective

The objective of this report is to perform a comprehensive packet-level analysis of the traffic.pcapng network capture using **Wireshark** and its command-line counterpart, **TShark**. The report addresses twenty predefined questions related to network traffic, including protocol distribution, IP and MAC address identification, and communication characteristics. Each answer is derived directly from the packet capture using appropriate display filters or TShark commands, ensuring that the analysis is accurate, reproducible, and verifiable..

Tool & Environment

Tool	Version	Purpose
Wireshark / TShark	4.2.2	Packet dissection, display filtering, protocol analysis
capinfos	4.2.2	Capture metadata — packet count, duration, timestamps
Bash / command line	4.1	Running filters and recording exact output for each question

Question-by-Question Analysis

E1. How many DHCP messages are present in the capture?

Filter: dhcp

Answer: **5**

Five bootp/dhcp frames were seen — one DHCP Release and a Discover/Offer/Request/ACK cycle for a single lease.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "dhcp" -T fields -e frame.number -e dhcp.option.dhcp
1336 7   (Release)      1474 1 (Discover)   1475 2 (Offer)
1476 3   (Request)      1477 5 (ACK)
$ tshark -r traffic.pcapng -Y "dhcp" | wc -l  -> 5
```

E2. How many ARP messages are present in the capture?

Filter: arp

Answer: **680**

Combined ARP requests and replies exchanged across the local subnet.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "arp" | wc -l
680
```

E3. Which IP address accessed baidu.com?

Filter: dns.qry.name contains "baidu"

Answer: 10.103.51.159

This host generated a long series of DNS lookups for baidu.com and its subdomains (www, sp0-sp2, baike, help, etc.), resolved via internal DNS server 10.103.0.20.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y 'dns.qry.name contains "baidu"' -T fields -e frame.number -e ip.src -e dns.qry.name
20754  10.103.51.159  www.baidu.com
20823  10.103.51.159  sp0.baidu.com
20894  10.103.51.159  baike.baidu.com  ... (20+ baidu subdomains total)
```

E4. How many packets were sent by source IP 10.103.0.20?

Filter: `ip.src==10.103.0.20`

Answer: 469

10.103.0.20 acts as the internal DNS resolver, so these are mostly DNS response packets.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "ip.src==10.103.0.20" | wc -l
469
```

E5. How many UDP packets exist in the capture?

Filter: `udp`

Answer: 9,102

UDP traffic is dominated by DNS, NBNS, SSDP, mDNS, LLMNR and QUIC.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "udp" | wc -l
9102
```

E6. How many SMB packets are in the capture?

Filter: `smb || smb2`

Answer: 518

Traffic rides over NetBIOS Datagram service (mailslot / NETLOGON / browser announcements), typical of a Windows domain LAN.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "smb || smb2" | wc -l
518
```

E7. How many packets were sent to destination IP 10.103.0.20?

Filter: `ip.dst==10.103.0.20`

Answer: 547

Client DNS queries and other service requests directed at the resolver.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "ip.dst==10.103.0.20" | wc -l
547
```

E8. Isolating IPv4 traffic only, what is the packet count?

Filter: `ip`

Answer: 24,906

IPv4 makes up the vast majority of the 26,456 total packets; the remainder is ARP and non-IP link-layer traffic.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "ip" | wc -l
24906
```

E9. What is the source IP of the last SMB packet?

Filter: `smb || smb2 (last frame)`

Answer: **10.103.50.202**

Frame 26452 — a NETLOGON broadcast to the subnet broadcast address 10.103.255.255.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "smb || smb2" -T fields -e frame.number -e ip.src -e ip.dst |
tail -1
26452  10.103.50.202  10.103.255.255
```

E10. What is the MAC address of IP 151.139.128.14?

Filter: `ip.addr==151.139.128.14`

Answer: **00:1c:7f:6c:96:3f**

This is the gateway-side MAC seen in every packet to/from this off-subnet host (OUI registered to Check Point Software Technologies).

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "ip.addr==151.139.128.14" -T fields -e eth.src -e eth.dst
10:60:4b:78:cd:4a  ->  00:1c:7f:6c:96:3f
00:1c:7f:6c:96:3f  ->  10:60:4b:78:cd:4a
```

E11. Which protocol is used in packet number 416?

Filter: `frame.number==416`

Answer: **DNS**

A DNS query carried over UDP, from 10.103.51.159 to the resolver 10.103.0.20.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "frame.number==416" -T fields -e ip.src -e ip.dst -e
_ws.col.Protocol
10.103.51.159  10.103.0.20  DNS
```

E12. What is the destination IP address in packet 416?

Filter: `frame.number==416`

Answer: **10.103.0.20**

Confirms packet 416 is a DNS query sent to the internal resolver.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "frame.number==416" -T fields -e ip.dst
10.103.0.20
```

E13. What is the source IP address of the first HTTP packet?

Filter: `http`

Answer: **10.103.51.159**

Frame 19007 — the first application-layer HTTP request in the capture, to 151.139.128.14 on TCP port 80.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "http" -T fields -e frame.number -e ip.src -e ip.dst | head -1
19007  10.103.51.159  151.139.128.14
```

E14. What is the source port of the first HTTP packet?

Filter: `http → tcp.srcport`

Answer: 64079

Client ephemeral port 64079 talking to server port 80.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "http" -T fields -e tcp.srcport -e tcp.dstport | head -1
64079 80
```

E15. What is the duration of the capture, in seconds?

Filter: `capinfos`

Answer: 348.11 sec (≈ 5 min 48 sec)

First packet at 12:53:50 UTC, last packet at 12:59:38 UTC on 2021-07-26.

Verification (TShark output):

```
$ capinfos traffic.pcapng
Number of packets: 26456
Capture duration: 348.109916 seconds
```

E16. How many NBNS packets are in the capture?

Filter: `nbns`

Answer: 963

NetBIOS Name Service broadcasts — routine on a Windows LAN using NetBIOS name resolution.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "nbns" | wc -l
963
```

E17. How many TCP packets have source port 443?

Filter: `tcp.srcport==443`

Answer: 7,275

Inbound HTTPS responses arriving from external/internet TLS servers.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "tcp.srcport==443" | wc -l
7275
```

E18. How many TCP packets have destination port 443?

Filter: `tcp.dstport==443`

Answer: 5,966

Outbound HTTPS requests initiated by LAN clients.

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "tcp.dstport==443" | wc -l
5966
```

E19. How many packets were sent to destination IP 204.79.197.200?

Filter: `ip.dst==204.79.197.200`

Answer: 116

204.79.197.200 is a Microsoft-owned address block (Bing/MSN infrastructure).

Verification (TShark output):

```
$ tshark -r traffic.pcapng -Y "ip.dst==204.79.197.200" | wc -l
```

E20. What is the MAC address of IP 204.79.197.200, and which vendor does it belong to?

Filter: `ip.addr==204.79.197.200`

Answer: 00:1c:7f:6c:96:3f — Check Point Software Technologies

Same gateway MAC as E10 — every off-subnet destination is reached through this Check Point firewall/gateway.:

Summary Table

S.no.	Question	Filter	Answer
E1	How many DHCP messages are present in the capture?	dhcp	5
E2	How many ARP messages are present in the capture?	arp	680
E3	Which IP address accessed baidu.com?	dns.qry.name contains "baidu"	10.103.51.159
E4	How many packets were sent by source IP 10.103.0.20?	ip.src==10.103.0.20	469
E5	How many UDP packets exist in the capture?	udp	9,102
E6	How many SMB packets are in the capture?	smb smb2	518
E7	How many packets were sent to destination IP 10.103.0.20?	ip.dst==10.103.0.20	547
E8	Isolating IPv4 traffic only, what is the packet count?	ip	24,906
E9	What is the source IP of the last SMB packet?	smb smb2 (last frame)	10.103.50.202
E10	What is the MAC address of IP 151.139.128.14?	ip.addr==151.139.128.14	00:1c:7f:6c:96:3f
E11	Which protocol is used in packet number 416?	frame.number==416	DNS
E12	What is the destination IP address in packet 416?	frame.number==416	10.103.0.20
E13	What is the source IP address of the first HTTP packet?	http	10.103.51.159
E14	What is the source port of the first HTTP packet?	http → tcp.srcport	64079
E15	What is the duration of the capture, in seconds?	capinfos	348.11 sec (≈ 5 min 48 sec)
E16	How many NBNS packets are in the capture?	nbns	963
E17	How many TCP packets have source port 443?	tcp.srcport==443	7,275
E18	How many TCP packets have destination port 443?	tcp.dstport==443	5,966
E19	How many packets were sent to destination IP 204.79.197.200?	ip.dst==204.79.197.200	116
E20	What is the MAC address of IP 204.79.197.200, and which vendor does it belong to?	ip.addr==204.79.197.200	00:1c:7f:6c:96:3f — Check Point Software Technologies

Conclusion

All twenty questions were answered exclusively using packet-level evidence from the traffic.pcapng capture, with reproducible TShark display filters and Wireshark analysis. No assumptions or extrapolated values were used, ensuring the accuracy, reliability, and verifiability of the results.

The analysis indicates that the captured network traffic originates from a Windows-based local area network (LAN) operating behind a Check Point gateway. The traffic consists of DNS queries, NetBIOS/SMB communication for local network and domain services, and predominantly TLS-encrypted web traffic. The capture also reveals an extended web browsing session to **baidu.com** initiated by the host **10.103.51.159**, demonstrating typical user web activity alongside routine network communication. Overall, the analysis provides a clear understanding of the network's communication patterns, protocol usage, and host interactions based solely on the captured packet data.